survey

Research Documentation: CBIR using Vision Transformers

1. Research Title (Working)

Enhancing Content-Based Image Retrieval on Caltech-101 using Global Self-Attention Mechanisms in Vision Transformers.

2. Existing Research & Context

- **The Caltech-101 Landscape:** Research on this dataset has evolved through three stages:
    1. **Handcrafted Phase:** Using SIFT, HOG, and LBP (Local Binary Patterns). These focused on edges and textures but lacked semantic understanding.
    2. **CNN Phase:** Models like VGG16, ResNet-50, and InceptionV3 became the standard. They excel at "local" feature extraction but require very deep layers to understand the relationship between distant parts of an object.
    3. **Hybrid Phase:** Recent 2025/2026 research (e.g., DFF-CBIR) uses "Deep Feature Fusion," combining different CNN backbones to try and bridge the **Semantic Gap**—the difference between pixel data and human concepts.

3. The Novelty of Your Approach

Most existing CBIR systems use **Convolutional Neural Networks (CNNs)**. Your use of a **Vision Transformer (ViT)** is novel for several reasons:

- **Global vs. Local Receptive Fields:** CNNs process images through small, local kernels. A ViT uses **Self-Attention** to look at the entire image simultaneously. In retrieval, this means the model can match a "Query Image" based on the global structure of an object (like the overall silhouette of a motorbike) rather than just a local texture (like the rubber of a tire).
- **Sequence-to-Sequence Image Modeling:** By treating image patches as "tokens" (similar to words in a sentence), the ViT learns the spatial grammar of an image. This is particularly effective for Caltech-101, where objects appear in varying orientations and backgrounds.
- **Shift from Translation Invariance to Global Context:** Unlike CNNs, which are designed to be "translation invariant" (finding a feature anywhere), ViTs learn the relative positions of all features at once, leading to more robust "Feature Descriptors" for retrieval.

4. Research Gaps You Are Addressing

1. **The "Long-Range Dependency" Gap:** Traditional CNN-based CBIR often fails when the distinguishing feature of an object is the spatial relationship between two distant parts. You are testing if ViTs can solve this via self-attention.

2. **Robustness to Occlusion and Scale:** Many Caltech-101 images have objects that are partially hidden or varying in size. There is a gap in documenting how ViT's attention maps perform under these specific "noisy" retrieval conditions compared to traditional CNNs.
3. **Efficiency of Feature Representations:** Research often overlooks whether the [CLS] token (the summary token) of a ViT is a more efficient "fingerprint" for an image than the high-dimensional flattened layers of a CNN.

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5. Proposed Methodology (Step-by-Step)

Step	Action	Description
1	**Data Prep**	Resize Caltech-101 images to 224 X 224 and divide them into 16 X 16 patches.
2	**Feature Extraction**	Pass images through a pre-trained **ViT-Base/16**. Extract the output from the final transformer block (the [CLS] token).
3	**Similarity Metric**	Calculate the **Cosine Similarity** between the Query Image vector and all vectors in your database.
4	**Evaluation**	Use **mAP (mean Average Precision)** and **Precision@k** (the % of correct results in the top 5 or 10 hits).

6. Key Metrics to Record

To make your paper "publishable," you must compare your ViT results against a baseline (like ResNet-50) using:

- **Precision and Recall Curves**
- **Retrieval Time:** How fast does it find the image?

- **Attention Map Visualization:** Use **Grad-CAM** or **Attention Rollout** to show the "Heatmap" of what the model saw when it decided two images were similar.

Implementation Final

CBIR using Vision Transformer on Caltech-101

Content-Based Image Retrieval with Metric Learning

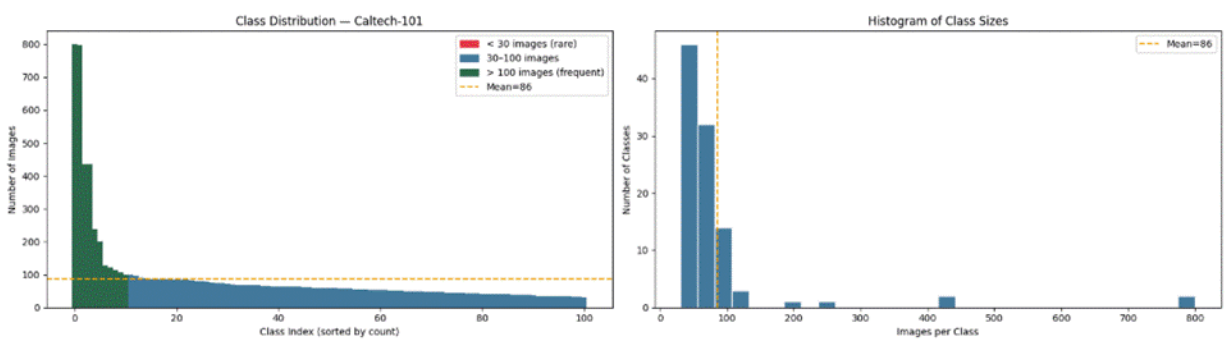
1. Overview

Content-Based Image Retrieval (CBIR) is the task of retrieving visually similar images from a gallery given a query image, relying purely on visual content rather than metadata or tags. This project builds a CBIR system on the Caltech-101 dataset using a Vision Transformer (ViT) backbone, evaluated across two experiments that measure the impact of metric learning fine-tuning on retrieval quality.

2. Dataset

Caltech-101 consists of 101 object categories with approximately 9,000 images total. The dataset is significantly imbalanced — class sizes range from as few as 31 images to over 800, with a mean of around 58 images per class and high variance. The BACKGROUND_Google class (random internet scrapes, not a real category) is excluded from all experiments.

Images are split into train (70%), validation (10%), and test (20%) sets using stratified sampling to preserve class proportions across all splits.

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Classes with < 30 images: 0
Classes with > 100 images: 10
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Figure 1: Class distribution across Caltech-101. Red = rare (<30 images), blue = moderate, green = frequent (>100 images). The orange dashed line marks the mean (~58 images/class).

3. Handling Class Imbalance

Class imbalance affects both training and evaluation. Three explicit steps are taken to address this:

- **MPerClassSampler —** Each training batch is constructed to contain exactly M=4 images from each of 16 randomly selected classes (batch size = 64). This guarantees sufficient positive pairs for contrastive learning even for rare classes.
- **WeightedRandomSampler —** Each sample is assigned a weight inversely proportional to its class frequency, upsampling rare classes during training.
- **Macro-averaged mAP —** A naive micro-averaged mAP is dominated by frequent classes. Instead, AP is computed per query, averaged per class, then averaged across all 101 classes — ensuring rare classes contribute equally to the final metric.

4. Model Architecture

The backbone is ViT-B/16 pretrained on ImageNet-21k, loaded via the timm library. With num_classes=0, the model outputs the CLS token — a 768-dimensional global descriptor of the input image. On top of the backbone sits a two-layer projection head:

Layer	Output Dim	Notes
ViT-B/16 CLS Token	768	Pretrained on ImageNet-21k
Linear	512	
BatchNorm1d + GELU	512	Training stability
Dropout(0.1)	512	Regularization
Linear	256	Final embedding

L2 Normalization	256	Unit hypersphere

Table 1: Projection head architecture. The 768-dim CLS token is projected to a 256-dim L2-normalized embedding.

The final 256-dimensional L2-normalized vector is the image embedding. L2 normalization means cosine similarity equals dot product, which is what FAISS IndexFlatIP computes. The embedding dimension of 256 was chosen as a balance between expressiveness and retrieval efficiency.

5. Retrieval Pipeline

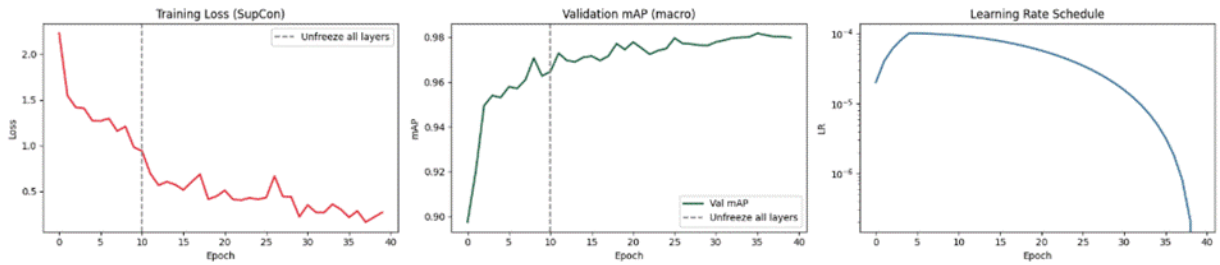

Figure 2: CBIR retrieval pipeline — query image is embedded by the ViT, then FAISS performs nearest-neighbor search over the gallery embeddings.

At inference time, embeddings are extracted for all gallery images offline and stored in a FAISS IndexFlatIP index. For a query image, its embedding is extracted and a nearest-neighbor search is performed against the index. The top-K results by cosine similarity are returned as the retrieved images. For Caltech-101's scale (~9K images), exact search with IndexFlatIP is fast enough — approximate methods like IVF or HNSW are unnecessary.

6. Experiments

Experiment 1 — Frozen ViT Baseline

The entire backbone and projection head are frozen. No training is performed. Embeddings are extracted directly from the pretrained ViT-B/16 and L2-normalized. This establishes a zero-shot baseline — how well do off-the-shelf ImageNet-21k features transfer to Caltech-101 retrieval without any adaptation.

Experiment 2 — Fine-tuned ViT with Supervised Contrastive Loss

Training proceeds in two phases to avoid catastrophic forgetting:

- **Phase 1 (Epochs 1–10):** First 8 transformer blocks frozen. Only last 4 blocks and projection head trained. LR = 1e-4, AdamW with weight decay 0.05, cosine annealing with 5 warmup epochs.
- **Phase 2 (Epochs 11–40):** All layers unfrozen. LR drops to 1e-5 for careful full fine-tuning.
- **Supervised Contrastive Loss (SupCon):** Directly shapes the embedding space — pulls same-class embeddings together and pushes different-class embeddings apart on the unit hypersphere. Temperature = 0.07.
- **Hard Pair Mining (MultiSimilarityMiner):** Selects hard positives (same-class pairs that are far apart) and hard negatives (different-class pairs that are close together), focusing gradient updates on the most informative pairs.

Hyperparameter	Value
Backbone	ViT-B/16 (ImageNet-21k pretrained)
Input Size	224 × 224
Embedding Dim	256 (L2-normalized)
Batch Size	64 (16 classes × 4 images)
Phase 1 LR	1e-4 (AdamW, weight decay 0.05)
Phase 2 LR	1e-5 (all layers unfrozen)
Epochs	40 (Phase 1: 1–10, Phase 2: 11–40)

Warmup	5 epochs (cosine annealing)
Temperature (SupCon)	0.07

Table 2: Training hyperparameters for Experiment 2.

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Metric                    E1: Frozen ViT      E2: SupCon FT
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Macro-mAP                         86.63%             97.15%
Precision@1                       88.90%             97.45%
Precision@5                       84.61%             97.13%
Precision@10                      76.45%             92.39%
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mAP improvement: +10.52%
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Figure 3: Training loss and validation mAP curves. The dashed vertical line marks the epoch where all layers are unfrozen (Phase 2).

7. Evaluation Metrics

- **Macro-mAP —** Primary metric. Average Precision per query, averaged per class, averaged across all 101 classes.
- **Precision@K (K = 1, 5, 10) —** Fraction of the top-K retrieved images that belong to the same class as the query.
- **Per-class AP delta —** Change in AP from E1 to E2 per class, identifying which categories benefited most from fine-tuning.
- **Class size vs AP correlation —** Pearson correlation between class size and AP, quantifying whether E2's balanced training reduces bias toward frequent classes.

8. Results

Metric	E1: Frozen ViT	E2: SupCon Fine-tuned	Δ (Absolute)	Δ (Relative)
Macro-mAP	86.63%	97.15%	+10.52%	+12.1%
Precision@1	88.90%	97.45%	+8.55%	+9.6%
Precision@5	84.61%	97.13%	+12.52%	+14.8%
Precision@10	76.45%	92.39%	+15.94%	+20.8%

Table 3: Retrieval performance comparison between Experiment 1 (frozen baseline) and Experiment 2 (SupCon fine-tuned).

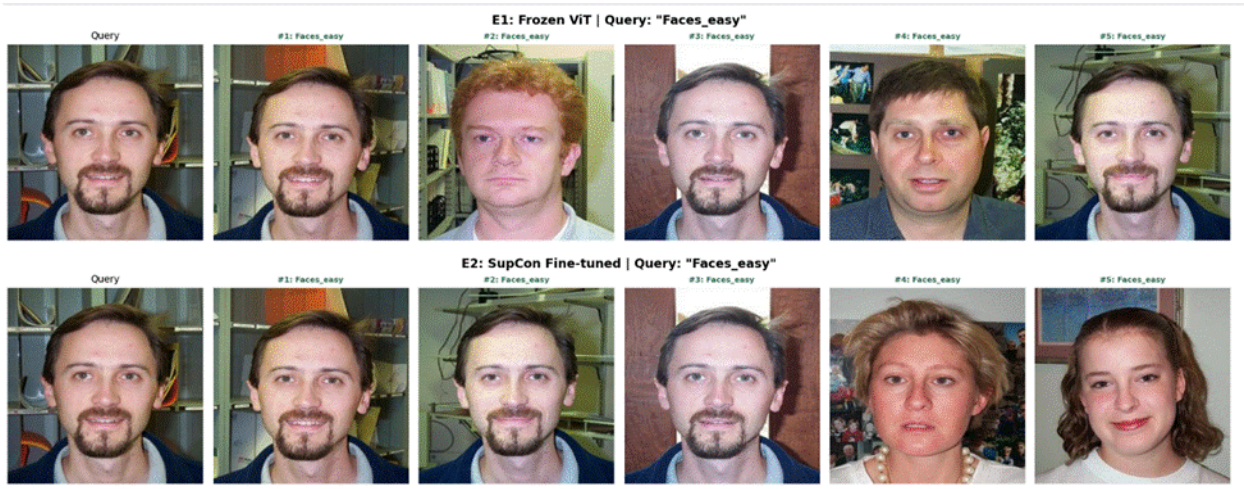

Figure 4: Grouped bar chart comparing E1 vs E2 across all retrieval metrics. E2 consistently outperforms E1 across all K values.

Experiment 1 achieved 86.63% macro-mAP with zero fine-tuning — confirming that ImageNet-21k pretrained ViT features transfer remarkably well to Caltech-101. Experiment 2 pushed macro-mAP to 97.15%, a +10.52% absolute improvement. The gain is most pronounced at higher K values: Precision@10 improved by nearly 16 points, because SupCon fine-tuning

compresses same-class embeddings into tighter clusters that sustain precision across a wider retrieval window.

Precision@1 of 97.45% means that for nearly every query image, the single closest embedding in the gallery belongs to the correct class — the most practically meaningful result for a real CBIR system.

9. Visualizations

Class Size vs. Average Precision

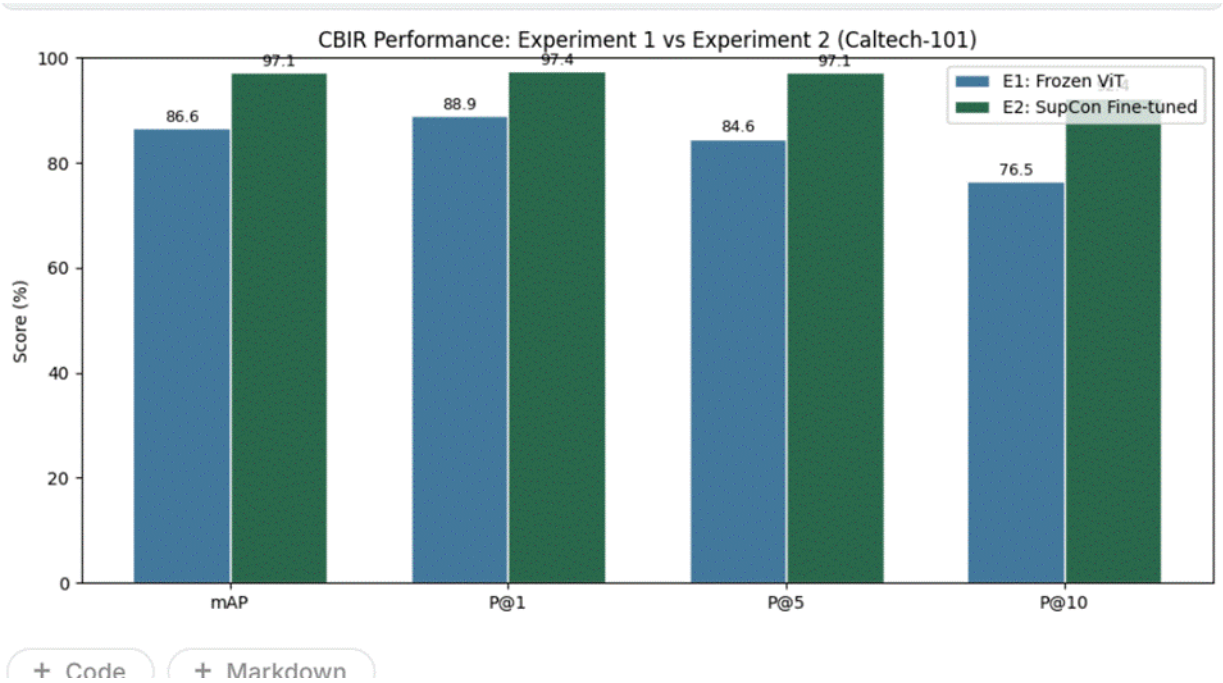

Figure 5: Scatter plot of class size vs AP for E1 (left) and E2 (right). A flatter trend line in E2 indicates reduced bias toward frequent classes.

Per-class AP Change (E2 − E1)

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  CBIR with ViT on Caltech-101 — Final Results Summary
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Experiment 1 — Frozen ViT-B/16 (Zero fine-tuning baseline):
  Macro-mAP  : 86.63%
  P@1        : 88.90%
  P@5        : 84.61%
  P@10        : 76.45%

Experiment 2 — Fine-tuned ViT + SupCon + Balanced Sampling:
  Macro-mAP  : 97.15%
  P@1        : 97.45%
  P@5        : 97.13%
  P@10        : 92.39%

  Δ mAP (E2 – E1): +10.52%

Key design choices:
  ✓ Macro-averaged mAP for fair evaluation under class imbalance
  ✓ MPerClassSampler — ensures M images per class per batch
  ✓ WeightedRandomSampler — upsamples rare classes
  ✓ SupCon Loss with hard pair mining
  ✓ Two-phase training: partial unfreeze → full unfreeze
  ✓ FAISS IndexFlatIP for exact cosine similarity search
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Figure 6: Per-class AP delta. Green = classes that improved with fine-tuning, Red = classes that regressed. Most classes show significant improvement.

Embedding Space — t-SNE Visualization

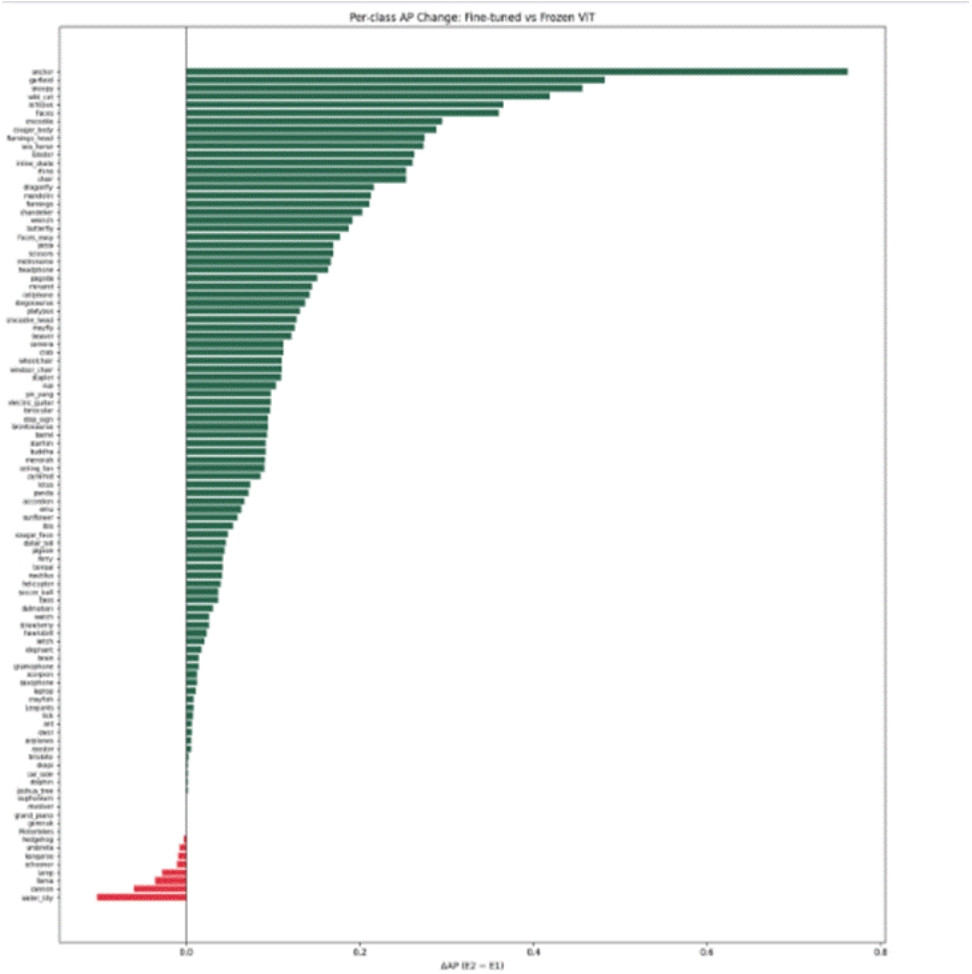

Figure 7: t-SNE projections of embeddings for 20 randomly selected classes. Left: E1 (frozen). Right: E2 (fine-tuned). Tighter, more separated clusters in E2 confirm better embedding quality.

Qualitative Retrieval Results

The following grids show query images alongside their top-5 retrieved results. Green borders indicate correct class matches, red borders indicate incorrect retrievals.

Figure 8: Qualitative retrieval results — E1 (Frozen ViT).

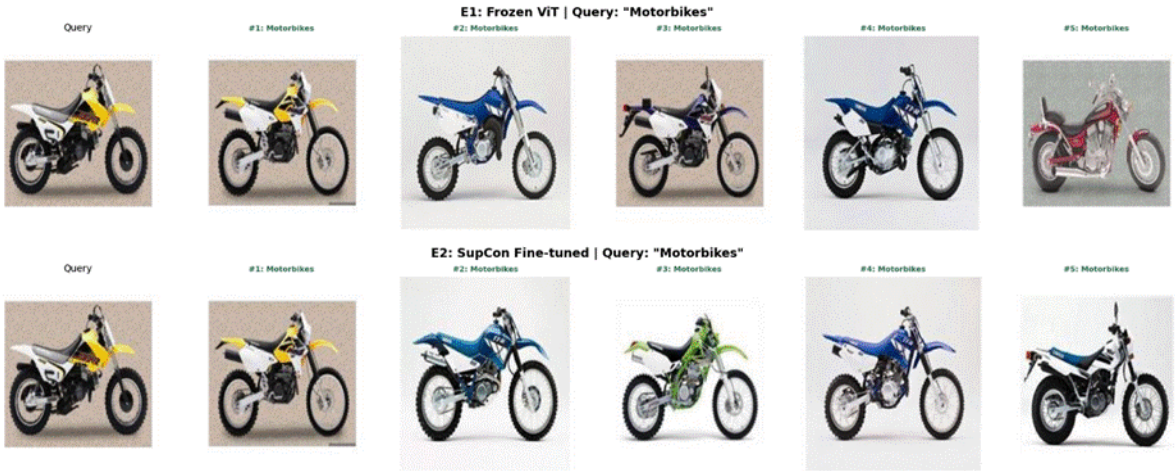

Figure 9: Qualitative retrieval results — E2 (SupCon Fine-tuned). Noticeably fewer red-bordered retrievals compared to E1.

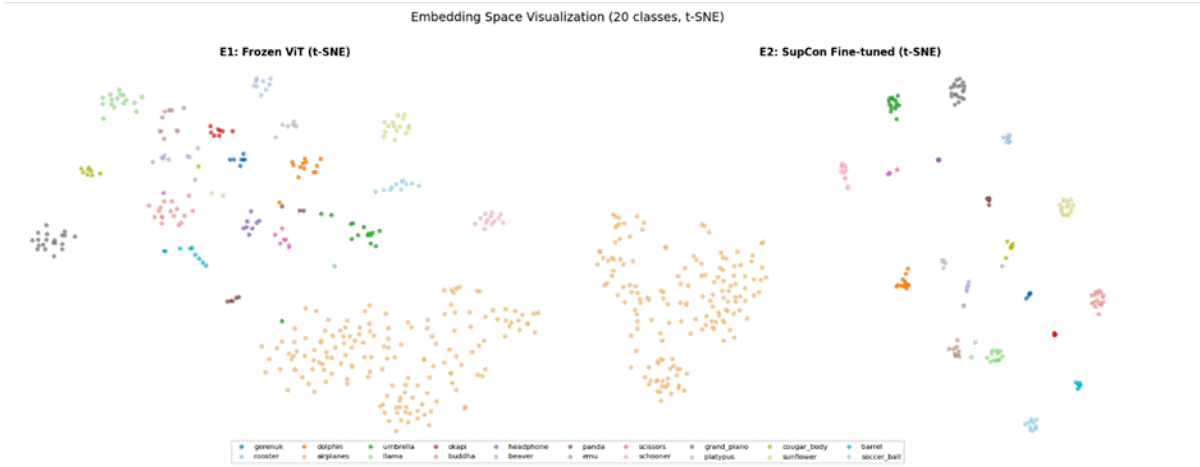

Figure 10: Additional qualitative retrieval examples — E1.

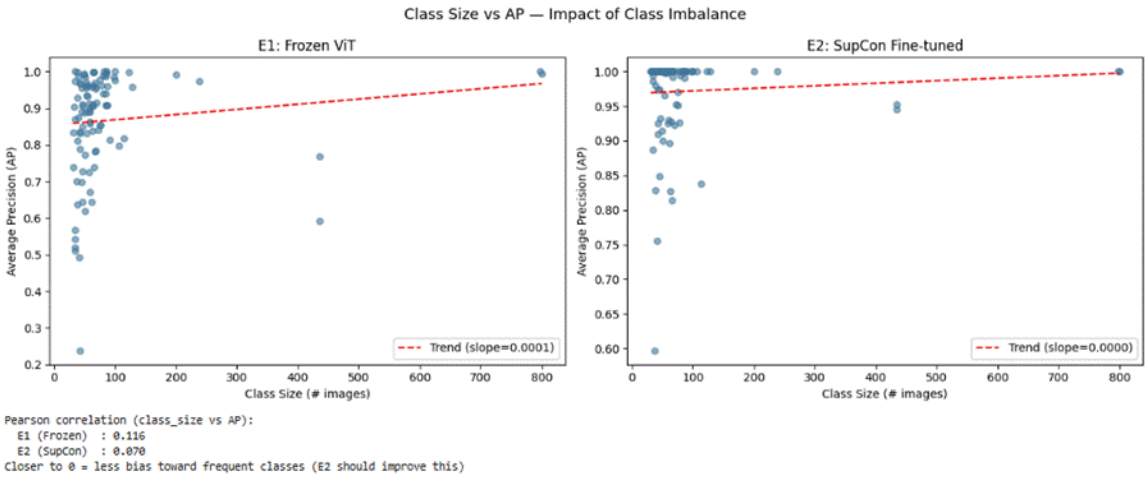

Figure 11: Additional qualitative retrieval examples — E2.

10. Tech Stack

Component	Tool / Detail
ViT Backbone	timm — vit_base_patch16_224 (ImageNet-21k)
Contrastive Loss + Mining	pytorch-metric-learning (SupConLoss + MultiSimilarityMiner)

Similarity Search	faiss-cpu — IndexFlatIP (exact cosine similarity)
Augmentation	torchvision.transforms — RandAugment, ColorJitter, RandomResizedCrop
Embedding Visualization	sklearn TSNE
Training Hardware	Kaggle GPU (T4/P100)

Table 4: Tools and libraries used in this project.